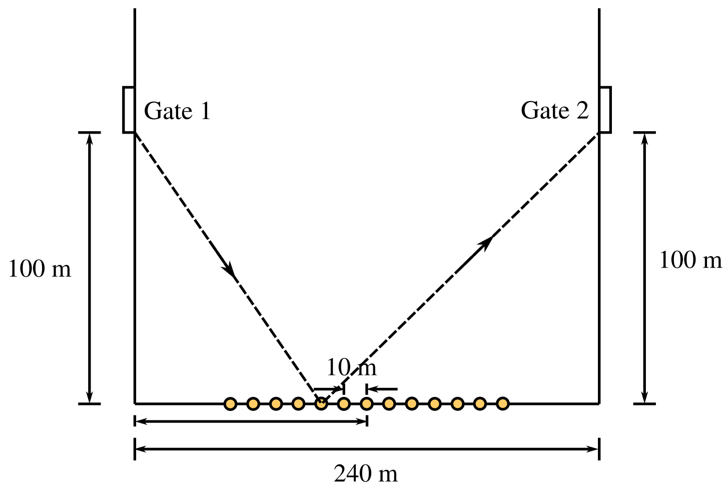
Worksheet: Optimal running paths

Task 1

Scenario: a marathon runner runs along some road and enters a rectangular field through a gate (Gate 1) and leaves through another 240m away (Gate 2). The runner has the option to grab a water bottle on 13 tables situated on a line on the southern edge 100 m away from both gates:



The 13 tables are at 60, 70, ... , 180 m away from the west edge along the southern edge of the field as shown in the picture.

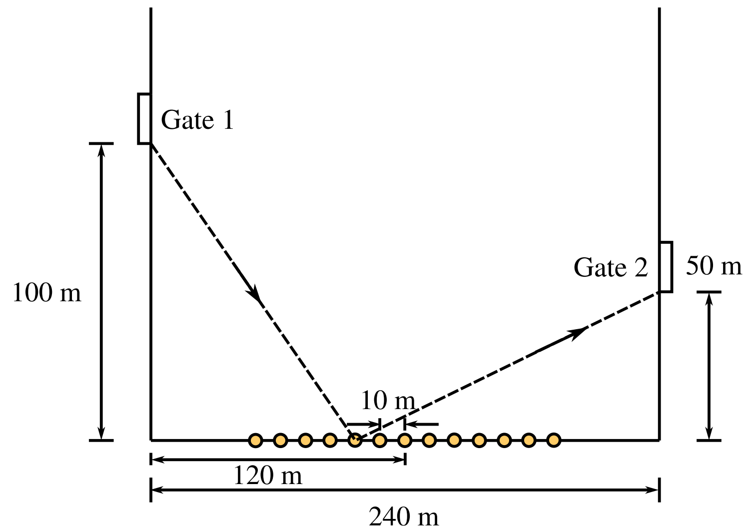
1. Pick a table. Write down expressions of the total distance of the runner's path in the field as he enters Gate 1, runs toward your choice of table then leaves through Gate 2, always traveling along straight lines as shown in the picture. Evaluate the expression to compute the total distance.
2. Do so for all 13 tables and fill in the numerical values to 2 decimal places. What are the changes you need to make to your expressions?

Table at...	60m	70m	80m	90m	100m	110m	120m
Tot. Dist.							
Table at...	130m	140m	150m	160m	170m	180m	
Tot. Dist.							

3. Graph the data points (total distance vs table choice) using Excel. Which table should the runner choose to run the least distance and save energy for the rest of the marathon?
4. (Extension) Is there a way you can arrive at this conclusion by computing the total distance for only 2 tables instead of all 13?

Task 2

Scenario: as in Task 1, south edge length is still 240m, but this time Gate 2 is now at 50 m away from the south edge not 100m. The location of Gate 1 is unchanged as in the picture below.



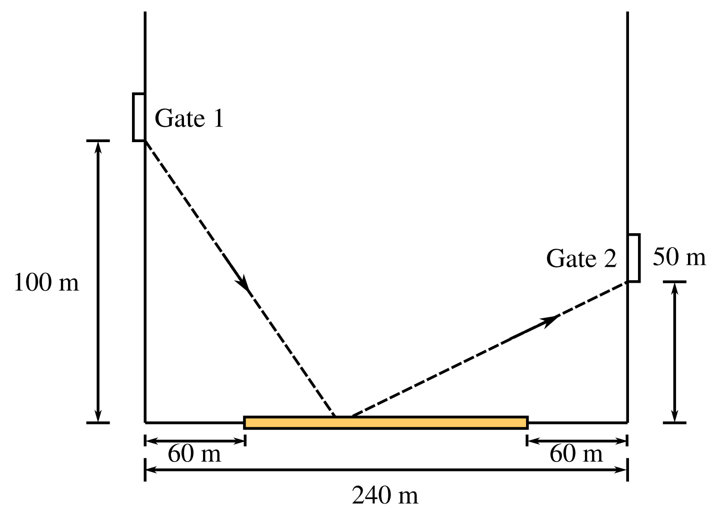
1. Before doing any calculation, write down your guess for which table might be the best choice and how/if you think it differs from the case for Task 1.

2. Repeat the calculation in Task 1 for Task 2: compute total distance for the running path for scenario in Task 2 and fill in the table below, then graph using Excel. Which table is the optimal choice this time? Does it confirm or disagree with your guess?

Table at...	60m	70m	80m	90m	100m	110m	120m
Tot. Dist.							
Table at...	130m	140m	150m	160m	170m	180m	
Tot. Dist.							

Task 3

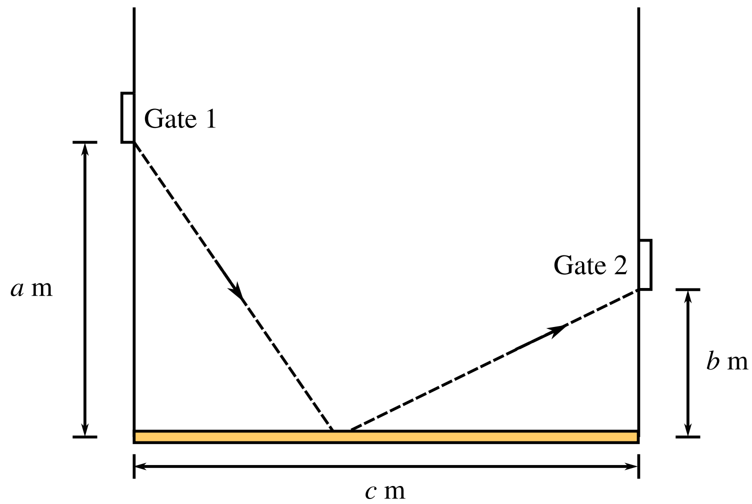
Scenario: the field and location of Gates 1 and 2 are identical to Task 2, but this time instead of 13 separate tables, there are a number of tables connected together to form a single long surface on which there are enough water bottles for the runners to grab. One end of the long table is 60 m away from the west edge, the other end is 60 m away from the east edge. No matter where the runner reaches the long table, he/she can grab a water and continue the marathon as shown below:



1. The total distance of the runner now varies continuously with different choices of the location on the long table where the runner chooses to grab the bottle. Set up an independent variable and write down the expression of the total distance in terms of this independent variable.
2. Find the optimal location which allows the runner to run the minimal distance between the two gates and grab a bottle from the long table. What operation do you need to perform to the function in order to find this optimal value of your independent variable? Why?
3. (Extension) Compare your optimal position to the position of the optimal table in Task 2. Is there any difference? If so, how do you make sense of this difference?

Task 4

Scenario similar to Task 3, but this time the table is as long as the southern edge. No actual number is given for the dimensions in this picture below, instead we view a, b, c as known constants.



1. Set up an expression for the total distance and an independent variable to determine the location on the long table where the runner grabs water bottle.
2. Find a simplified formula that provides the optimal position on the long table to minimise runner total distance on the field for any positive number choices for a, b, and c.
3. (Extension) Is there a way we can arrive at this formula without using calculus?